

Retail Sales Performance Analytics

End-to-End Data Analysis: Python → Power BI

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June – September 2024

Raw Data → Python → Power BI → Business Insights

The Business Problem

The business observed a sharp revenue slowdown across Q3 2024.

71%

**Revenue decline
June → September**

01 — WHERE?

Where is revenue
coming from?

02 — WHY?

Why is revenue
declining?

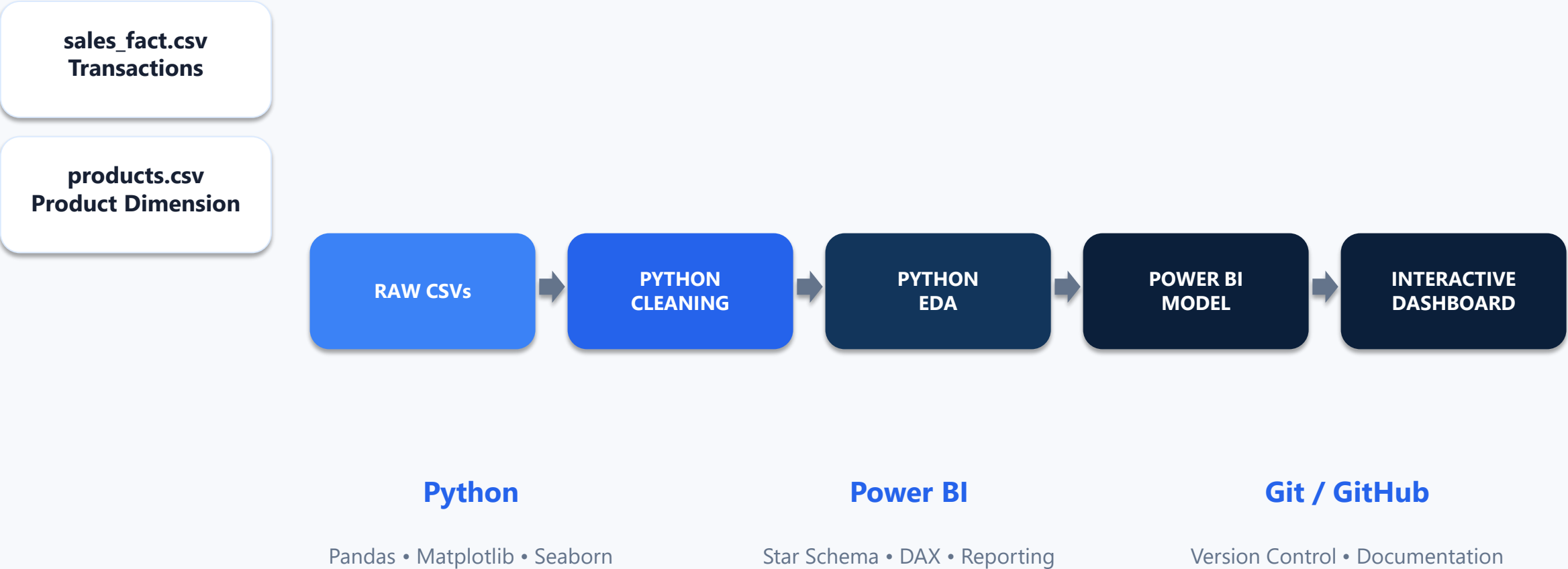
03 — WHAT NEXT?

What should the
business do
next?

60 clean transactions → 3 data-backed actions

From Raw Data to Business Insight

Every insight is traceable through a structured analytical pipeline.



Data Cleaning — Turning Raw Data into Reliable Data

The analysis started with imperfect raw data and ended with a validated dataset.

RAW DATA

61

Raw Transactions

- Missing quantities
- Missing unit prices
- 1 duplicate transaction
- Inconsistent region text
- 1 negative-quantity error



CLEAN DATA

60

Clean Analysis-Ready Records

- 0 Missing Values
- 0 Duplicates
- Consistent Formatting
- Fully Validated Dataset

Headline KPIs

Four months of performance at a glance

₹1.50M

Total Sales

₹606.18K

Total Profit

60

Total Transactions

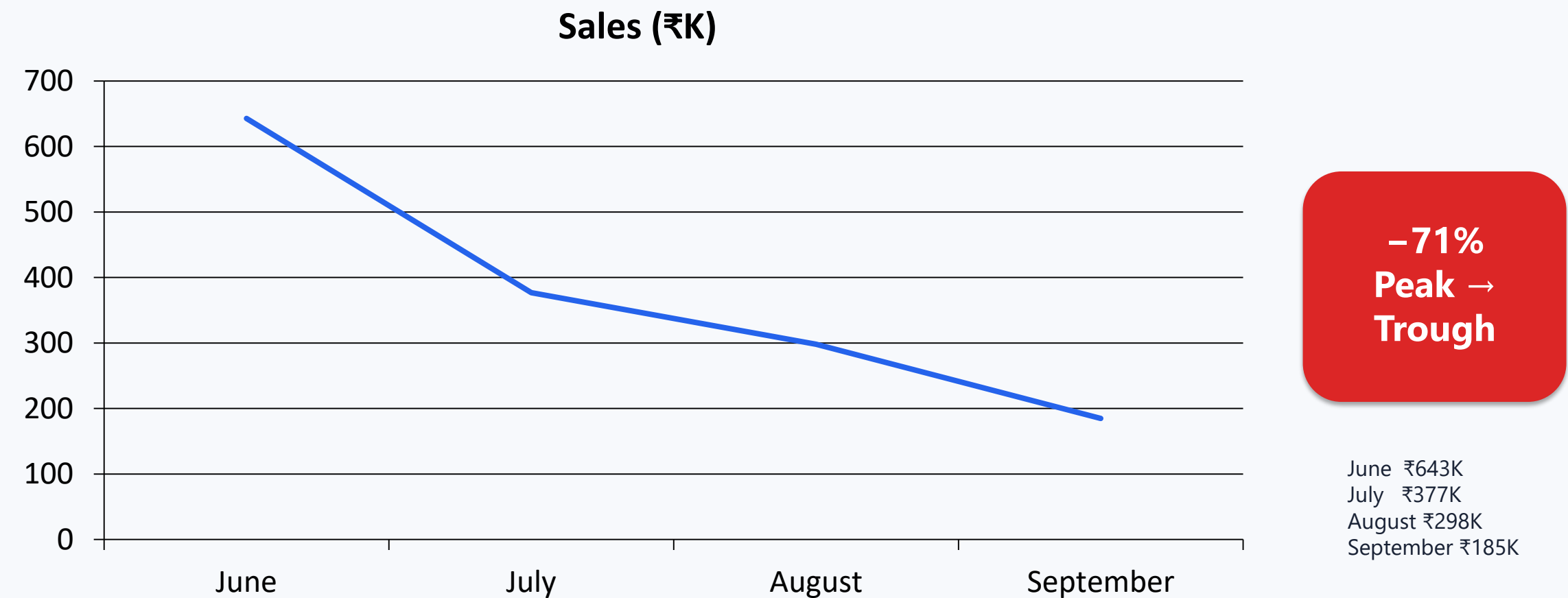
40.36%

Profit Margin

Healthy unit economics — the key issue is demand, not profitability.

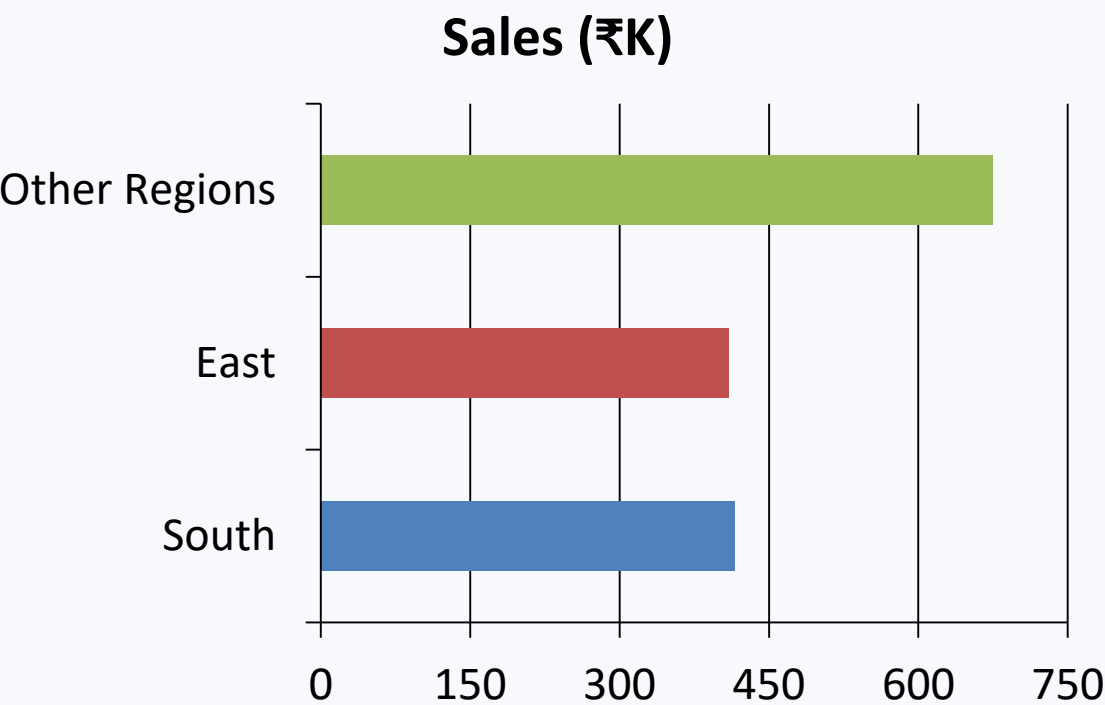
The Core Problem — Revenue Is Declining Fast

Revenue declined every month from June through September.

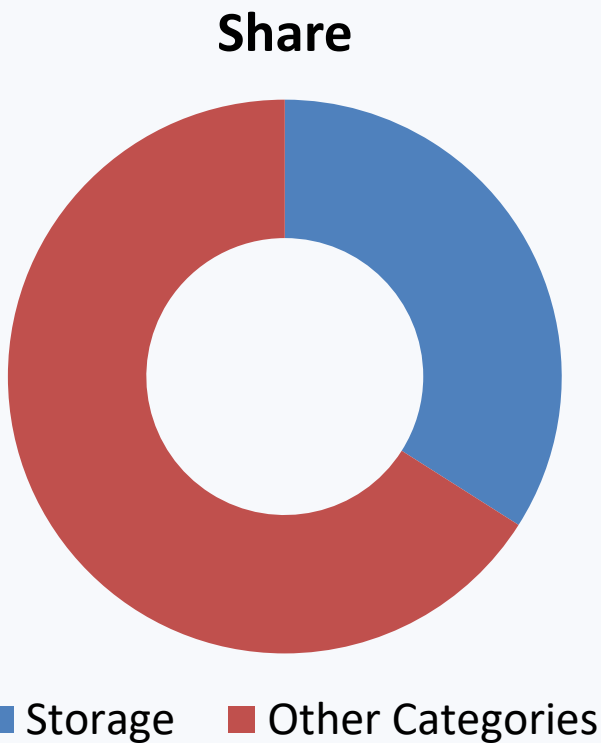


Where the Revenue Is Coming From

Strength is concentrated in specific markets and products.



South + East = ₹825K → 55% of total sales



Storage = 34% of total sales

A Hidden Payment Method Gap

Credit Card performance trails the other major payment methods.

CREDIT CARD

₹295K

~27% gap

Structural issue
worth investigating

Other Payment Methods

UPI

₹398K–₹405K

Debit Card

₹398K–₹405K

Cash on Delivery

₹398K–₹405K

Three channels perform in a similar range; Credit Card is the clear outlier.

Three Actions for Q4

Turn the findings into focused business action.

01 **Recover the Revenue Trend**

Launch a Q4 retention & recovery campaign.

First diagnose whether the decline is volume-driven or value-driven before committing budget.

URGENT

02 **Double Down on Proven Strength**

Prioritize inventory and marketing spend on Storage × South/East.

Protect the strongest existing revenue engine instead of spreading resources evenly.

HIGH

03 **Fix the Credit Card Gap**

Audit the Credit Card checkout experience.

Investigate payment failures, UX friction, authorization issues and customer behavior.

TARGETED

Thank You

Questions?

Full analysis, code and dashboard available in the project repository.

Python

Data Cleaning + EDA

Power BI

Interactive Dashboard

GitHub

Code + Documentation